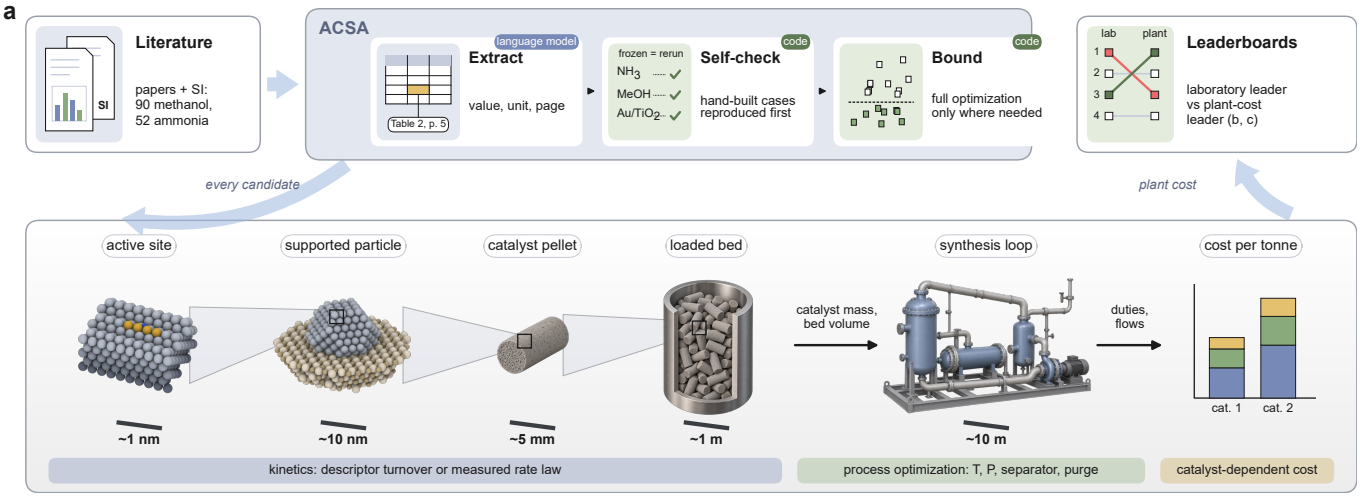
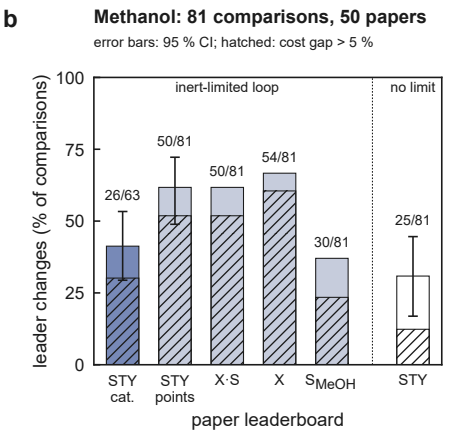
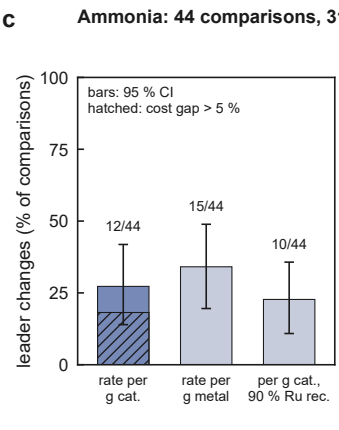
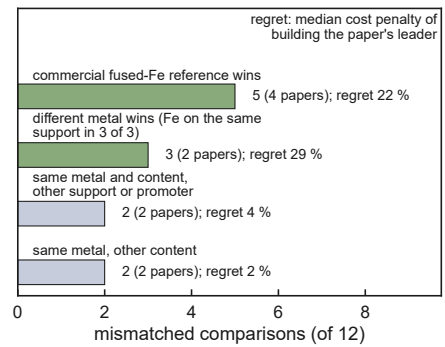
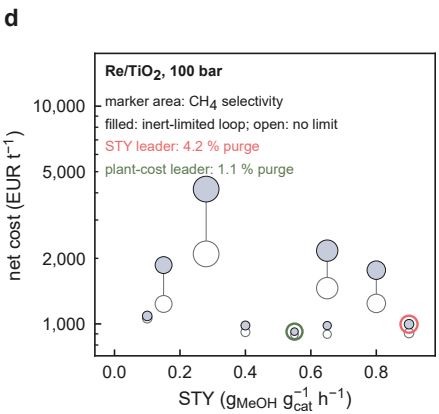
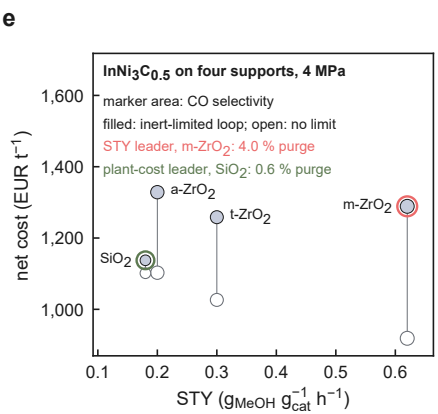

a
Literature
SI
papers + SI:
90 methanol,
52 ammonia
ACSA
language model
Extract
Table 2, p. 5
value, unit, page
frozen = rerun
NH3
MeOH
Au/TiO2
Self-check
code
hand-built cases
reproduced first
Bound
code
full optimization
only where needed
lab
plant
Leaderboards
laboratory leader
vs plant-cost
leader (b, c)
every candidate
plant cost
active site
supported particle
catalyst pellet
loaded bed
synthesis loop
cost per tonne
~1 nm
~10 nm
~5 mm
~1 m
catalyst mass,
bed volume
duties,
flows
~10 m
cat. 1
cat. 2
kinetics: descriptor turnover or measured rate law
process optimization: T, P, separator, purge
catalyst-dependent cost

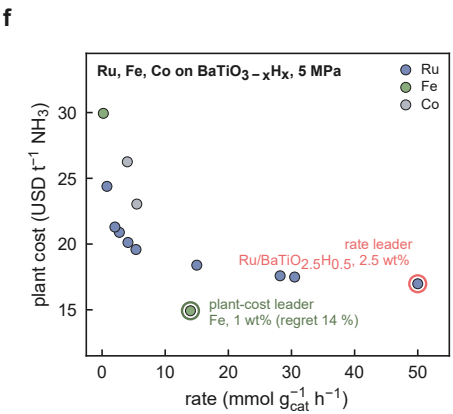

b
Methanol: 81 comparisons, 50 papers
error bars: 95 % CI; hatched: cost gap > 5 %
inert-limited loop
no limit
leader changes (% of comparisons)
26/63
50/81
50/81
54/81
30/81
25/81
STY cat.
STY points
X·S
X
SMeOH
STY
paper leaderboard


c
Ammonia: 44 comparisons, 31 papers
bars: 95 % CI
hatched: cost gap > 5 %
leader changes (% of comparisons)
12/44
15/44
10/44
rate per g cat.
rate per g metal
per g cat., 90 % Ru rec.


regret: median cost penalty of building the paper's leader
commercial fused-Fe reference wins
5 (4 papers); regret 22 %
different metal wins (Fe on the same support in 3 of 3)
3 (2 papers); regret 29 %
same metal and content, other support or promoter
2 (2 papers); regret 4 %
same metal, other content
2 (2 papers); regret 2 %
mismatched comparisons (of 12)


d
Re/TiO2, 100 bar
marker area: CH4 selectivity
filled: inert-limited loop; open: no limit
STY leader: 4.2 % purge
plant-cost leader: 1.1 % purge
net cost (EUR t−1)
STY (gMeOH gcat−1 h−1)


e
InNi3C0.5 on four supports, 4 MPa
marker area: CO selectivity
filled: inert-limited loop; open: no limit
STY leader, m-ZrO2: 4.0 % purge
plant-cost leader, SiO2: 0.6 % purge
a-ZrO2
t-ZrO2
m-ZrO2
SiO2
net cost (EUR t−1)
STY (gMeOH gcat−1 h−1)


f
Ru, Fe, Co on BaTiO3−xHx, 5 MPa
Ru
Fe
Co
rate leader
Ru/BaTiO2.5H0.5, 2.5 wt%
plant-cost leader
Fe, 1 wt% (regret 14 %)
plant cost (USD t−1 NH3)
rate (mmol gcat−1 h−1)